```
>g++ MatrixWorkbench.cpp -o MatrixWorkbench.exe

>MatrixWorkbench
*************** Matrix Workbench ***************
1. Create Matrix A
2. Create Matrix B
3. Display the current matrices
4. Check multiplication compatibility
5. Perform A x B
6. Perform B x A
7. Multiply a matrix by a scalar
8. Restore the original matrices
0. Exit
Choice: 1
Enter number of rows for Matrix A (1-10): 3
Enter number of columns for Matrix A (1-10): 3
Choose input method:
1- Manual
2- Random
Choice: 2
Enter minimum random value: 1
Enter maximum random value: 99

Great! Matrix A has been created.
Matrix A ID: 19784
Order: 3 x 3
   83   80   42
   69   91   24
   43   75   30

Press any key to return to Main Menu!
*************** Matrix Workbench ***************
1. Create Matrix A
2. Create Matrix B
3. Display the current matrices
4. Check multiplication compatibility
5. Perform A x B
6. Perform B x A
7. Multiply a matrix by a scalar
8. Restore the original matrices
0. Exit
```

```
Choice: 2
Enter number of rows for Matrix B (1-10): 3
Enter number of columns for Matrix B (1-10): 3
Choose input method:
1- Manual
2- Random
Choice: 1
Enter B[0][0]: 1 2 3
Enter B[0][1]: Enter B[0][2]: Enter B[1][0]: 4 5 6
Enter B[1][1]: Enter B[1][2]: Enter B[2][0]: 7 8 9
Enter B[2][1]: Enter B[2][2]:
Great! Matrix B has been created.
Matrix B ID: 18129
Order: 3 x 3
   1   2   3
   4   5   6
   7   8   9

Press any key to return to Main Menu!
*************** Matrix Workbench ***************
1. Create Matrix A
2. Create Matrix B
3. Display the current matrices
4. Check multiplication compatibility
5. Perform A x B
6. Perform B x A
7. Multiply a matrix by a scalar
8. Restore the original matrices
0. Exit
Choice: 3
------ Matrix A ------
Matrix ID: 19784
Order: 3 x 3
  83  80  42
  69  91  24
  43  75  30

------ Matrix B ------
Matrix ID: 18129
Order: 3 x 3
   1   2   3
   4   5   6
```

7 8 9

Press any key to return to Main Menu!
*************** Matrix Workbench ***************
1. Create Matrix A
2. Create Matrix B
3. Display the current matrices
4. Check multiplication compatibility
5. Perform A x B
6. Perform B x A
7. Multiply a matrix by a scalar
8. Restore the original matrices
0. Exit
Choice: 4
------ Compatibility Report ------
Matrix A Order: 3 x 3
Matrix B Order: 3 x 3
A x B: Valid
Result Order: 3 x 3
B x A: Valid
Result Order: 3 x 3

Note: Even when both products are defined, matrix multiplication is generally not commutative.

Theoretical Complexity:
Time Complexity: O(1)
Auxiliary Space: O(1)

Press any key to return to Main Menu!
*************** Matrix Workbench ***************
1. Create Matrix A
2. Create Matrix B
3. Display the current matrices
4. Check multiplication compatibility
5. Perform A x B
6. Perform B x A
7. Multiply a matrix by a scalar
8. Restore the original matrices
0. Exit
Choice: 5
------ Matrix Multiplication Report ------

```
Matrix 1:
   83   80   42
   69   91   24
   43   75   30

Matrix 2:
    1    2    3
    4    5    6
    7    8    9

A x B:
  697  902 1107
  601  785  969
  553  701  849

Result Order: 3 x 3
Scalar Multiplications: 27
Additions: 18

Theoretical Complexity:
Time Complexity: O(rA x cA x cB)
For square matrices n x n: O(n^3)
Auxiliary Space: O(rA x cB)
------------------------------------------

Press any key to return to Main Menu!
*************** Matrix Workbench ***************
1. Create Matrix A
2. Create Matrix B
3. Display the current matrices
4. Check multiplication compatibility
5. Perform A x B
6. Perform B x A
7. Multiply a matrix by a scalar
8. Restore the original matrices
0. Exit
Choice: 6
------ Matrix Multiplication Report ------

Matrix 1:
    1    2    3
```

```
    4    5    6
    7    8    9

Matrix 2:
   83   80   42
   69   91   24
   43   75   30

B x A:
  350   487   180
  935  1225   468
 1520  1963   756

Result Order: 3 x 3
Scalar Multiplications: 27
Additions: 18

Theoretical Complexity:
Time Complexity: O(rA x cA x cB)
For square matrices n x n: O(n^3)
Auxiliary Space: O(rA x cB)
------------------------------------------

Press any key to return to Main Menu!
*************** Matrix Workbench ***************
1. Create Matrix A
2. Create Matrix B
3. Display the current matrices
4. Check multiplication compatibility
5. Perform A x B
6. Perform B x A
7. Multiply a matrix by a scalar
8. Restore the original matrices
0. Exit
Choice: 7
Select matrix:
1- Matrix A
2- Matrix B
Choice: 2
Enter scalar value: 99

Before Scalar Multiplication:
```

```
    1    2    3
    4    5    6
    7    8    9
```

After Scalar Multiplication:

```
   99  198  297
  396  495  594
  693  792  891
```

Scalar Multiplications: 9
Writes: 9

Theoretical Complexity:
Time Complexity: O(r x c)
Auxiliary Space: O(1)

Press any key to return to Main Menu!
*************** Matrix Workbench ***************
1. Create Matrix A
2. Create Matrix B
3. Display the current matrices
4. Check multiplication compatibility
5. Perform A x B
6. Perform B x A
7. Multiply a matrix by a scalar
8. Restore the original matrices
0. Exit
Choice: 8

Matrix A Restored:

```
   83   80   42
   69   91   24
   43   75   30
```

Matrix B Restored:

```
    1    2    3
    4    5    6
    7    8    9
```

Original matrices restored successfully!

Press any key to return to Main Menu!

*************** Matrix Workbench ***************
1. Create Matrix A
2. Create Matrix B
3. Display the current matrices
4. Check multiplication compatibility
5. Perform A x B
6. Perform B x A
7. Multiply a matrix by a scalar
8. Restore the original matrices
0. Exit
Choice: 0

Thank you for using the Matrix Workbench! Goodbye!

C:\Users\Gargi\MatrixWorkbench>